

Yinjie Zhao

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🐙 [YinjieZhao22](https://github.com/YinjieZhao22)

🌐 [Website](#)

EDUCATION

University of Michigan, Ann Arbor, MI, United States
B.S.E. in Data Science

Expected May 2026

Shanghai Jiao Tong University, Shanghai, China
B.S.E. in Mechanical Engineering

Expected Aug 2026

RESEARCH INTERESTS

My research interest involves building reliable and efficient systems for machine learning.

PUBLICATIONS

OpGuard: Bitwise Alignment for Precise and General Debugging of Production LLM Training

Ziming Zhou, **Yinjie Zhao**, Hang Zhu, Wenxiao Wang, Zhihao Bai, Yun Zhang, Shuguang Wang, Haibin Lin, Peng Huang
OSDI 2026

EXPERIENCE

University of Michigan – Order Lab

Ann Arbor, MI, United States

Undergraduate Research Intern – Supervisor: *Prof.* Ryan Huang

May 2025 – Present

- **OpGuard: LLM Training Debugging System**

- Designed and implemented a tensor parallel (TP) simulator to emulate single-GPU behavior on multi-GPU setups, mitigating the numerical difference and enabling bitwise comparison across executions with different TP configurations.
- Applied OpGuard to diagnose 10 open-source issues on Github, including DeepSpeed, Megatron-LM, and transformers.

- **TrainCheck: Training with Confidence** [\[repo\]](#) (OSDI 2025)

- Designed and implemented the runtime checker with a watchdog to monitor the trace file changes.
- Designed a new proxy to achieve `torch.compile` /distributed training compatibility and reduce runtime overhead.
- Revised the invariant relations (infer/checking).

Shanghai Jiao Tong University – Digital Media and Computer Vision Lab

Shanghai, China

Undergraduate Research Intern – Supervisor: *Prof.* Ran Yi

Feb. 2024 – Mar. 2025

- **Video Customization with Subject and Motion Customization**

- Designed an alternating strategy to finetune DreamBooth at the video level for subject identity and train a motion LoRA by finetuning only Q, K projection weights within AnimateDiff for motion learning.
- Designed a dynamic scaling of spatial and temporal LoRA to synthesize personalized videos.

SELECTED PROJECTS

Simulated Basic Operating System *EECS 482 (Introduction to Operating Systems)*

- Built a multicore thread library, a virtual memory pager (COW, eviction), and a concurrent Network File System.

HONORS & AWARDS

- Yuliming Scholarship December 2023
- Outstanding Undergraduate Scholarship (B Level), SJTU December 2023

SKILLS

- Technical: C++/C, Python, CUDA, HTML, JavaScript, SQL, MATLAB, LaTeX, Linux, R
- ML: PyTorch (torch.compile), DeepSpeed, Megatron-LM, Hugging Face Transformers, vLLM
- Languages: English (TOEFL 105/120), Chinese (Native)

TEACHING

- Teaching Assistant — ME2400J (Intro to Dynamics and Vibrations), SJTU Summer 2024

LEADERSHIP & SERVICE

- Minister of the Outer Liaison Department, UM-SJTU Joint Institute Student Union Sep. 2022 – Jul. 2024